Defense Script

Percolation Analysis of Vein Reticulation as a Structural Basis for Damage Tolerance in Leaf Venation Networks under Simulated Vascular Damage

Nicko Fajardo | spoken runtime about fifteen minutes | ten blocks

Read the grey text; say the black text. Board cues tell you where to point, never what to read aloud. Every number here traces to Part 8 of the study mastery guide.

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1. The cold open 45 sec

BOARD → Cuscuta photograph, full bleed, no text

This is *Cuscuta*. Dodder. It is a holoparasite, which means it gave up on photosynthesis entirely. No chlorophyll. Barely any roots. No real leaves.

It survives by finding another plant, wrapping around its stem, and drilling a feeding structure called a haustorium straight into the host's plumbing. And then it drinks.

It is one of about four thousand parasitic plant species, and it is the one that is best studied, which is why it is on the screen. But the question I started with is the general one: how much water does a parasitic sink have to steal before it kills the host's vascular system?

Slow, low, and let the pause before the last sentence sit. This is the only stretch of the defense where you have their attention without earning it, so do not spend it on the title of your study. Greet the panel after this, or fold it into block 2. Name the organism here because a named organism is what makes an opening image work, but name it as an exemplar, so that nothing later in the defense rests on it being that particular species.

2. The negative result, stated first 1:30

BOARD → The 2.6 percent tile

So I built it. I solved the entire leaf as an electrical circuit. Kirchhoff's laws, every vein a resistor, and the parasite entered as a leak conductance calibrated to withdraw twenty percent of the leaf's total water throughput. That is a serious sink, and I will say up front where the twenty percent comes from: it is a hemiparasite measurement, Jiang and colleagues in 2003, because no holoparasite water-flux fraction has ever been measured. So the number is not species-specific, and I do not model a species. I model a sink of that strength.

It did almost nothing.

It depressed the network's minimum water potential by 0.026 megapascals. That is 2.6 percent of the safety margin in my model. The other 97.4 percent is my own stipulated

boundary conditions, and I say so in the paper. To actually reach the modelled cavitation threshold you would need a host-parasite gradient of about six megapascals. The largest gradient anyone has measured in a real system is about half a megapascal.

So the hypothesis I started with was wrong by roughly an order of magnitude, and I have the number that kills it. But it left a much better question standing.

Deliver this completely flat. No apology, no nervous laugh, no hedging. You are reporting a measurement, and the calm is the whole point. Doing this in your first ninety seconds means everything after it inherits the credibility, and it permanently disarms the question about whether your parasite result is a failure, because you answered it first and on your own terms.

3. The real question 1:30

BOARD → Reticulation concept panel, branching against looped

Here is the question that survived. Forget who is doing the damage. How much damage can a leaf's vein network take at all, before it stops being one connected network?

To see why that is not obvious, think about two cities.

The first is a suburb. Every street is a cul-de-sac hanging off one main road. Close one street and every house past it is cut off, because there is no other way in.

The second is a grid. Every block connects to four others. Close a street and you go around.

Leaf veins are exactly this. Some leaves are suburbs, where veins branch outward and never rejoin. Most flowering plants are grids, where the veins loop back and reconnect, closing small islands called areoles. That looping is called reticulation, and it is my variable.

And we know it matters in real plants. Sack and colleagues cut a leaf's midrib, the main highway, and lost about 75 percent of its hydraulic conductance. Then they cut the small looped veins instead, and measured no detectable penalty at all. The grid routed around it.

What nobody had done was turn that into a number you can compare across species.

The sentence to land is "some leaves are suburbs, most are grids." If the panel remembers one thing about your project a week from now, it is that. Point left for the suburb and right for the grid, physically, and keep the sides consistent when the metaphor returns in block 9.

4. Percolation 1:00

BOARD → Progressive fragmentation montage

So keep closing streets in the grid city. At first nothing happens. Everyone still gets everywhere. Keep going, and still nothing. Still nothing.

And then, over a very narrow range, the map stops being one city and becomes a scatter of isolated towns. It does not degrade. It shatters.

That is percolation theory, and it is the right tool precisely because connectivity does not decay smoothly. It holds, and holds, and then it goes.

So each leaf becomes a graph. Nodes are vein junctions, edges are vein segments. I delete junctions at random and watch the largest surviving connected group, the giant connected component. My threshold is the fraction of junctions I have to remove before that giant component falls below half its original size, half the time. I call it the critical removed fraction.

If they interrupt to ask why half, that is engagement, not an attack. Answer in one breath: these networks span 421 nodes to 300,633, four and a half orders of magnitude, and a spanning-cluster criterion behaves differently at those two extremes where a half-size rule applies identically at every scale. Add that it is operational rather than a critical point in the physics sense, and that you re-ran everything against a source-anchored criterion as a check, which correlates at rho 0.912 with every association preserved.

5. The data, and the flaw you hand them 1:10

BOARD → Methodology panel, dataset and pipeline

The networks are real, not simulated. They come from the Matos 2024 Dryad deposit, 122 species from the UC Berkeley Botanical Garden, traced with a convolutional neural network that preserves the minor veins and the areoles. There are 120 usable networks, and my primary percolation cohort is 100 of them after a resolvability screen.

Now the thing you are going to find if I do not tell you. One hundred of those 120 files are calibrated crop windows, not entire leaves. Not one file in the deposit carries a petiole flag.

So I never say leaves. The title says leaf venation networks, every result says networks, and I pick my source node by a documented rule, the endpoint of the widest vein in the window, which is the hydraulically most proximal point where water enters. Does that turn a crop into a whole lamina? No, and I do not claim it does.

Volunteering this costs eight seconds and buys the rest of the defense. A panelist who finds a hidden limitation stops trusting everything else you said; a panelist who hears you name it first assumes you know everything else too.

6. The threshold 1:00

BOARD → Hero tile 0.1998, then the collapse curve

Across 100 real networks, the median critical removed fraction is 0.1998. Call it twenty percent of junctions. That is the shatter point.

That is one leaf's collapse curve. Two thousand four hundred Monte Carlo trials per network, fixed seeds, fully reproducible, and the threshold is the inflection point of a logistic fit.

I also ran a mode where busier junctions are likelier to fail. That drops the threshold to 0.1685, about 15.7 percent lower, and it did so in 100 out of 100 networks. Not 99. All of them.

One point of precision. That is not an attack. Removal is probabilistic and weighted, not rank-ordered, and nothing in biology computes a centrality measure. It is a bounding scenario: random removal gives the expected response, weighted removal gives its pessimistic edge.

7. Architecture predicts tolerance 1:25

BOARD → Threshold against predictors, then the quartile table

So does being a grid actually buy you anything?

Mean degree, the average number of vein segments meeting at a junction, which is my measure of how looped a network is, predicts the threshold at rho 0.733. Fractal dimension, which measures how thoroughly the venation fills space, comes second at 0.496.

Sort the cohort into quarters by how looped they are, and the median threshold climbs at every step, from 0.158 to 0.231. The most reticulate quarter tolerates 7.2 percentage points more damage than the least. That is a 45.7 percent relative gain.

Two caveats before you ask for them. First, fractal dimension is partly just measuring size: 73 percent of its variance is explained by network size. So I recomputed every association with log network size partialled out, and they all got stronger rather than weaker. Fractal dimension went from 0.496 to 0.633. A pure size artifact would have collapsed under that control. Second, my two-predictor model has an R-squared of 0.507, so roughly half the variance in the threshold sits outside anything I measured. I would rather say that than have you find it.

8. The intervention 2:35

BOARD → Per-network change after enhancement, the before-and-after plot

But everything so far is correlation. In real leaves, looping and spatial complexity travel together, so the most reticulate leaves also tend to be the most space-filling ones. Nature will not separate them for you.

So I did two things.

First, I built networks where I could separate them. A synthetic system in which connectivity and spatial complexity vary independently, which no real leaf will allow. Restricted to the range real leaves actually occupy, connectivity beats spatial complexity by about eight to one in standardised terms.

Second, and this is the part I would defend hardest, I intervened on the real leaves themselves. I added loops to 105 real leaves and re-measured each one.

I took each leaf's own junction coordinates and computed a Delaunay triangulation to propose candidate new veins. I chose Delaunay specifically because it is parameter-free. There is no knob, so there is nothing for a skeptic to accuse me of tuning in my own favour. I discarded the candidates that already existed and added the rest shortest-first, so the smallest open areoles close first, which is the most plausible biological order.

And I stopped at a mean degree of 2.811, which is the 90th percentile of the cohort's own connectivity. So no leaf in this experiment is ever made denser than the most reticulate leaf I actually observed. Nothing is extrapolated into an architecture that does not exist.

Then I re-measured, driving both versions with the same random number stream, so the Monte Carlo noise is shared between the two arms instead of adding into their difference. Only the edge set changed. Every leaf is its own control: same species, same size, same shape, same tracing quality, same phylogeny. Literally the same leaf.

Ninety-three of 105 got tougher. Twelve did not move. Zero got worse.

The median gain is 0.0095 in removable fraction, with a p-value on the order of ten to the minus eighteen, and it holds across six different tracing-quality subsets with no reversal in any of them. In the cleanest subset alone, the artifact-free grade-A tracings, thirteen improved, eight were unchanged, and none worsened.

This block is the study, and it runs 2:35, the longest in the script. Say "zero got worse" slower than everything around it and stop for one beat before continuing. Slides 12 and 13 both belong to this block.

9. The scope limit 1:45

BOARD → The three boxplots, dispersed against lesioned

Now the result that constrains everything I just told you.

Random removal scatters damage across the whole lamina. But a great deal of real damage is not scattered. A lesion, an insect bite, a necrotic patch is one solid region. So I re-ran everything removing contiguous patches at matched removal fractions.

Localised damage is far less dangerous. The threshold rises from 0.1998 to 0.4951 for a single lesion. That is two and a half times more damage tolerated, in 100 out of 100 networks, without a single exception.

And here is the part that matters. The reticulation relationship disappears. Under one lesion, the correlation between looping and tolerance falls from 0.737 to negative zero point zero zero zero. Statistically nothing.

Which makes sense if you go back to the city. A single lesion closes every street in one neighbourhood. The rest of the map is still contiguous, so you route around the hole, and the grid connections between neighbourhoods are never tested. Scattered damage is what attacks exactly those connections, and that is where reticulation earns its keep.

So my claim is scoped, deliberately. Reticulation governs tolerance to dispersed damage. That qualifier is load-bearing and it is on my board.

One more thing worth knowing. Fractal dimension behaves the opposite way. It is the only predictor that survives multi-lesion damage, at 0.523. Different damage geometry, different governing trait. That is a finding, not a hedge.

10. The close 2:05

BOARD → Conclusion panel, with the Cuscuta image returning small

So. Leaf venation networks lose global connectivity after roughly twenty percent of junctions fail. Reticulation raises that threshold: it predicts it across 100 species, it carries the effect when I decouple it experimentally, and when I add it to a leaf's own geometry the leaf gets tougher, in 93 of 105 cases, with none made worse. Conditional on the damage being dispersed.

And the parasite I started with is still in the paper, as the thing that disciplined it. It is the reason I never once describe my threshold as hydraulic collapse, because the percolation

engine never reads a conductance, a radius, or a water potential. Not once. It is purely topological, and I retitled the entire study when I realised that.

What I want next is the experiment I cannot do on a computer: real leaves, controlled damage, and a measurement of whether the threshold I calculate predicts where they actually fail.

And somebody needs to put a pressure chamber on a parasite and its host, because nobody has. I want to be exact about the scope of that claim, because it is the one place in this defense where naming the species matters. For *Cuscuta* specifically it is a published negative: the 2025 haustorium review by the field's own specialists states that no total water potential, osmotic potential, or turgor value has been reported for any *Cuscuta*-host interaction. Widened to holoparasites as a class it is my own search result rather than someone else's published statement: I found no holoparasite water-flux fraction, no haustorial conductance, and no host-parasite gradient of any kind. That is not a gap I invented for a future-work slide. I went looking for that number for months.

Do not let the two halves of that claim collapse into one. The Cuscuta version is a published negative you can attribute; the holoparasite version is a search result you can only vouch for yourself. Stating the narrower one first and then widening it is what keeps both defensible. Generalising it to "parasitic plants" would be simply false, because hemiparasite water relations have been measured, and the 20 percent figure in block 2 is one of those measurements.

The ending is earned because you opened on the parasite. Close the loop, then stop. Do not say thank you until after the silence.

Handling the panel

When they interrupt

Interruption is engagement, not hostility. The failure mode is losing your place and restarting from the top, which is why the script is built as ten blocks rather than one speech. After any interruption, say "coming back to it" and re-enter at the block you were in. Never re-summarise what you already said. If they cut you badly for time, the three blocks that carry the study are 2, 8 and 9; everything else is setup.

When a panelist starts lecturing

Let them. Take an actual note. Then: "That is useful, can I check I have it? You are saying this, and the place it bites me is this." You have turned a lecture into a collaboration in front of the rest of the panel.

When you do not know

You will get a question you cannot answer, and how you handle it is worth more than the answer. Say: "I do not know. Here is what I would need to find out," and name the specific measurement or analysis. "My instinct is this, but I have not tested it and I would not want to claim it." Never invent a number, never hedge into vagueness, and never use "that is beyond the scope" as a shield. Scope is a real answer only when you can say why the boundary sits where it does.

When a panelist is wrong

Do not roll over and do not fight. "I might be misunderstanding. My understanding is this, because of this. Is there a case where that breaks down?" Confident and coachable at once, which is the highest-scoring posture available in the room.

The questions that carry the most weight

The full question bank is in the earlier presentation script. These decide the grade. The bolded sentence in each answer does the work; everything after it is support you can drop if you are cut off.

Q. Which parasite are you actually modelling?

None of them, and the paper does not claim one. **The sink is a generic haustorial leak conductance, not a species.** The one number that fixes its strength, the twenty percent diversion, is a hemiparasite measurement from Jiang and colleagues in 2003, used because no holoparasite water-flux fraction has ever been measured for any species. *Cuscuta* appears in my introduction as motivation, and in my limitations as the scope of a published negative. It does not appear in my model.

And I would rather add the reason that distinction matters than have you find it. *Cuscuta* preferentially attaches to host stems, petioles and pedicels, not to leaf blades, and *Orobanche* attaches to roots. My networks are traced leaf-lamina windows. So a haustorial node placed inside a lamina is a modelling device, and I declare it as one rather than as an observed configuration.

Q. Correlation is not causation.

Agreed, which is why the correlation is not my main result. **I intervened.** Synthetic networks with the two axes decoupled by construction, and then loop addition on each real leaf's own coordinates, with that leaf as its own control. Species, size, shape, tracing quality and phylogeny are all held constant, because it is literally the same leaf. That is as close to an experiment as a computational study gets.

Q. You added veins for free. Real leaves have to pay for them.

That is the sharpest criticism of the intervention and I agree with it completely. **My model has no construction-cost, mechanical, or photosynthetic term**, so it identifies the direction of a relationship rather than an optimum. Real venation sits on a trade-off surface, which Ronellenfitsch and Katifori mapped, and greater hydraulic safety in real leaves comes with higher leaf mass per area and thicker conduit walls. Adding a cost term is the obvious next version.

Q. How can you claim hydraulic failure when you never measured conductivity or embolism?

I do not claim it, and I removed the claim myself. The percolation engine never reads a conductance, a radius, or a water potential. It is purely topological. I had been describing the threshold as hydraulic collapse, I checked the code, and it was not. So I retitled the study to a structural basis for damage tolerance and rewrote the claims. That was uncomfortable and it made the paper much stronger.

Q. So the parasite result is a failure.

It is a quantified negative, which is more useful than a vague positive. **At the calibrated strength the parasite contributes 2.6 percent of the reported margin, and reaching cavitation would need six megapascals against half a megapascal ever measured.** And I will volunteer the weakness inside my own negative: the claim that no vessels embolise is threshold-dependent, and it inverts at a minus one cavitation threshold. I swept it, and the onset gradient is 1.1 megapascals at minus one, 6.0 at minus two, and 9.5 at minus three. So the correct statement is that the absence of embolism is specific to thresholds at or below about minus 1.10 megapascals within the range I swept.

Q. Your degree-weighted removal only drops the threshold 15.7 percent. In a scale-free network, targeting hubs is catastrophic. Why is yours so mild?

Because leaf venation is not scale-free, and that is a structural finding rather than an excuse. **Planarity bounds node degree, and my maximum degree across the entire cohort is four to five.** There are no hubs to decapitate. That narrow degree distribution is precisely why weighted removal is a modest perturbation here and a catastrophe in a Barabasi-Albert network.

Discipline

The four numbers you must never fumble

Number	What it is
0.1998	median threshold, uniform removal, n = 100
93 of 105, zero worsened	the intervention
2.46 times	the localised-damage advantage
2.6 percent	the parasite's share of the modelled margin

Two sentences loaded at all times

> The percolation engine never reads a conductance, a radius, or a water potential. The threshold is purely topological.
>
> It only holds for dispersed damage. Under a single lesion the relationship disappears entirely.

Vocabulary

A panelist who knows the field will catch every one of these, and the substitutions are the difference between sounding careful and sounding overclaimed.

Never say	Always say
attack, targeted attack	degree-weighted or betweenness-weighted removal
hydraulic collapse	network fragmentation, structural collapse
validated	verified, for the resistance-column check
pre-symptomatic, early warning	sub-critical, pre-fragmentation, or nothing
conductance	modelled conductance
driven by the parasite	with the sink imposed as a boundary condition
reticulation improves robustness	raises tolerance to dispersed damage
proves, or shows that plants	demonstrates in this model that
leaves	leaf venation networks

The metaphor budget

Suburb against grid is the only metaphor in the script. Use it three times: block 3 to introduce reticulation, block 9 to explain why the correlation dies under a single lesion, and once more in questioning if someone looks lost. A fourth use makes it sound like the metaphor is all you have.

The last thing before you walk in

Do not reread the results. You know the results. Read Part 5 of the study mastery guide, the do-not-claim table. Restraint is what wins the room.